ModelIt!

Teacher's Guide

Sponge vs. Pavement

Where Does the Water Go When It Rains?

Quick Reference

Grade Level	Grade NE
NGSS Standard	MS-ESS2-4, MS-ESS3-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: MS-ESS2-4, MS-ESS3-3

Develop a model to describe the cycling of water through Earth's systems driven by energy from the sun and the force of gravity. Apply scientific principles to design a method for monitoring and minimizing a human impact on the environment.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how ground type and rainfall determine the split between absorption and runoff.
Disciplinary Core Idea	**ESS2.C The Roles of Water in Earth's Surface Processes** — Water cycles through Earth's systems — ground type determines whether precipitation is absorbed or becomes surface runoff.
Crosscutting Concept	**Cause and Effect** — Students identify that ground type and rainfall amount cause predictable effects on absorption and runoff patterns.

CAST Testing Alignment

MULTIPLE CHOICE:

A city replaces a grass field with a new parking lot. During the next heavy rain, what will MOST LIKELY happen? A) Less flooding nearby B) More flooding nearby C) No change in flooding D) The rain will stop sooner

CONSTRUCTED RESPONSE:

Your model shows that pavement creates much more runoff than soil during heavy rain. Using this evidence, explain why cities that have more pavement tend to have worse flooding problems. Use the words 'absorption' and 'runoff' in your answer.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Water is constantly cycling through Earth's systems — evaporating from oceans, forming clouds, falling as rain, soaking into soil or flowing across surfaces. Understanding how water moves across different surfaces is critical for managing floods, droughts, and water quality.

Permeable vs. Impermeable

Permeable surfaces have tiny spaces (pores) between particles that water can flow through. Soil, gravel, and sand are permeable. Impermeable surfaces block water — concrete, asphalt, and solid rock have no pores. When rain hits permeable ground, it soaks in. When it hits impermeable ground, it runs off.

The Urban Flooding Problem

In natural landscapes, about 10% of rainfall becomes runoff and 90% soaks in. In cities with 75% impermeable surfaces, those numbers flip — 55% or more becomes runoff. This overwhelms storm drains, causes flash flooding, and carries pollutants into waterways. Urban flooding is getting worse as cities grow.

Green Infrastructure Solutions

Cities are increasingly using 'green infrastructure' to reduce runoff: rain gardens, permeable pavement, green roofs, bioswales, and urban trees. These solutions mimic natural ground by allowing water to soak in. Some cities, like Philadelphia, have invested billions in green infrastructure instead of expanding traditional storm drains.

Connection to Beaver Ecology

In natural watersheds, beaver dams serve a similar function to green infrastructure — they slow water flow, spread it across floodplains, and allow it to soak into the ground. Research shows that watersheds with active beaver populations have significantly less flooding and better groundwater recharge than those without beavers.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Ground Type and Rainfall Amount as external components (conditions we set or observe) and Water Absorbed and Runoff as internal components (outcomes that change based on ground type and rainfall).

E **Establish Relationships**
Students determine that rainfall increases BOTH absorption and runoff, but ground type determines the RATIO — soil favors absorption while pavement favors runoff.

V **Visualize & Model**
Students build a model showing how two externals produce two competing internals, with ground type acting as a 'switch' that changes the balance.

E **Evaluate Outcomes**
Students run soil vs. pavement scenarios and light vs. heavy rain scenarios to see how the balance shifts.

R **Revise & Extend**
Students add slope angle or vegetation cover to see how additional factors modify the absorption-runoff split.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Have you ever noticed that some places get huge puddles when it rains and other places do not? Why is that?"*

Watch For

Show two photos side by side: a flooded parking lot vs. a forest after the same rainstorm. Ask: What is different?

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to build a model that explains WHERE water goes when it rains — and why cities flood."*

Watch For

Have students read objectives. Pre-teach 'permeable' and 'impermeable' with demo: pour water on a sponge vs. a plastic plate.

Slide 3: Big Question

3 min

Say: *"Think about it — the same rain falls on a forest AND a parking lot. One floods, one does not. What is going on?"*

Watch For

Have students discuss in pairs and share out their hypotheses.

Slide 4: LEVER Framework

2 min

Say: *"We are going to model this system to figure out exactly what controls where water goes."*

Watch For

Briefly review LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What factors determine where rain goes? What can we control, and what happens as a RESULT?"*

Watch For

Guide sorting: Ground Type and Rainfall are external. Water Absorbed and Runoff are internal. Key question: Can the same rain produce BOTH absorption and runoff?

Slide 6: Activity 2: Connections

8 min

Say: *"Here is what makes this interesting — the same rain can go two different directions depending on the ground!"*

Watch For

Draw arrows. Key teaching: Water Absorbed and Runoff are COMPETING — what soaks in cannot run off, and vice versa.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us test soil vs. pavement with the same heavy rain. Predict first — then run it!"*

Watch For

Run all three scenarios. Have students record predictions. Key moment: the dramatic difference between soil and pavement results.

Slide 8: Discoveries

5 min

Say: *"Now you know WHY cities flood! When we pave over natural ground, we turn rain into a problem."*

Watch For

Discuss: What percentage of our school grounds is paved? What does that mean for water?

Slide 9: STEM Challenge

12 min

Say: *"Let us put this to the test — we are going outside to map our school and predict where water goes!"*

Watch For

Distribute clipboards and mapping supplies. Go outside to map surface types and predict water flow.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Ground Type, Rainfall Amount
- **INTERNAL:** Water Absorbed, Runoff

Relationships

- **Ground Type to Water Absorbed:** VARIES BY TYPE — Soil (permeable) absorbs a large percentage of rainfall. Gravel absorbs a moderate amount. Pavement (impermeable) absorbs almost nothing. The type of ground acts like a switch that determines how much water soaks in.
- **Ground Type to Runoff:** VARIES BY TYPE (opposite of absorption) — Soil produces little runoff. Pavement produces heavy runoff. Whatever water does NOT absorb becomes runoff — so these two outputs are inversely related.
- **Rainfall Amount to Water Absorbed:** POSITIVE (+) — More rain means more water available to soak in — up to a limit. Eventually the ground becomes saturated and cannot absorb any more.
- **Rainfall Amount to Runoff:** POSITIVE (+) — More rain means more runoff, especially on impermeable surfaces. Even on soil, very heavy rain overwhelms the absorption rate and creates runoff.

Key Discoveries

- Soil absorbs much more water than pavement — it acts like a sponge
- Pavement is nearly impermeable — almost all rain becomes runoff
- More rain means more of BOTH absorption and runoff, but the ground type determines the split
- When cities replace soil with pavement, they create flooding problems because water has nowhere to soak in

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Water just disappears into the ground"	Water does not disappear — it moves through soil into underground aquifers (groundwater) or is absorbed by plant roots. The ground acts as a filter and a storage system. This groundwater slowly feeds streams, wells, and springs. When we pave over ground, we cut off this entire underground water system.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"All dirt absorbs the same amount of water"	Different soil types absorb very different amounts. Sandy soil is very permeable and absorbs quickly. Clay soil is much less permeable and can cause runoff even in natural areas. Healthy soil with lots of organic matter and earthworm tunnels absorbs the most.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Flooding is only caused by too much rain"	While heavy rain is one factor, flooding is heavily influenced by what the ground is made of. A moderate rain can cause flooding on pavement while the same rain on soil causes no flooding at all. Land use is often MORE important than rainfall amount in determining flood risk.